Printed Pages: 1 **IST Semester Test-2** **UID No:** 25MCI0099.

Academic year 2026 – 2027

Program Name/Code: Master of Computer Applications (Artificial Intelligence and Machine Learning)

Subject Code: 25CAH-723

Subject Title Internet of Things

Semester: 3

Time: 1 Hour **Maximum Marks: 20**

Instructions: Attempt all questions

Q. No	Statement	CO Mapping
Section A **5 x 2 = 10 marks**		
1	Motor-driven IC is used to run the DC motor while interfacing Arduino with the motor. Explain with reason.	CO1
2	PWM pins are different from digital pins. How?	CO1
3	Brief the role of delay function in the embedded programming.	CO2
4	Calculate the current (I) flowing from a conductor for a resistor value of 5.67 ohms and voltage of 3.3 Volts.	CO1
5	ISM bands are used for free communication like Bluetooth technology. Why?	CO1
Section B **2 x 5 = 10 marks**		
6	Interface an LED with Arduino UNO to generate the dimming effect or vary the intensity of the LED. Design a circuit and write its code.	CO3
7	Design a method to remove the bouncing problem of a pushbutton.	CO3